

Tutorial 1a

Hydrological Modelling – Development of a HEC-HMS model

HEC-HMS model development

Learning objectives

After completing this lab you will be able to

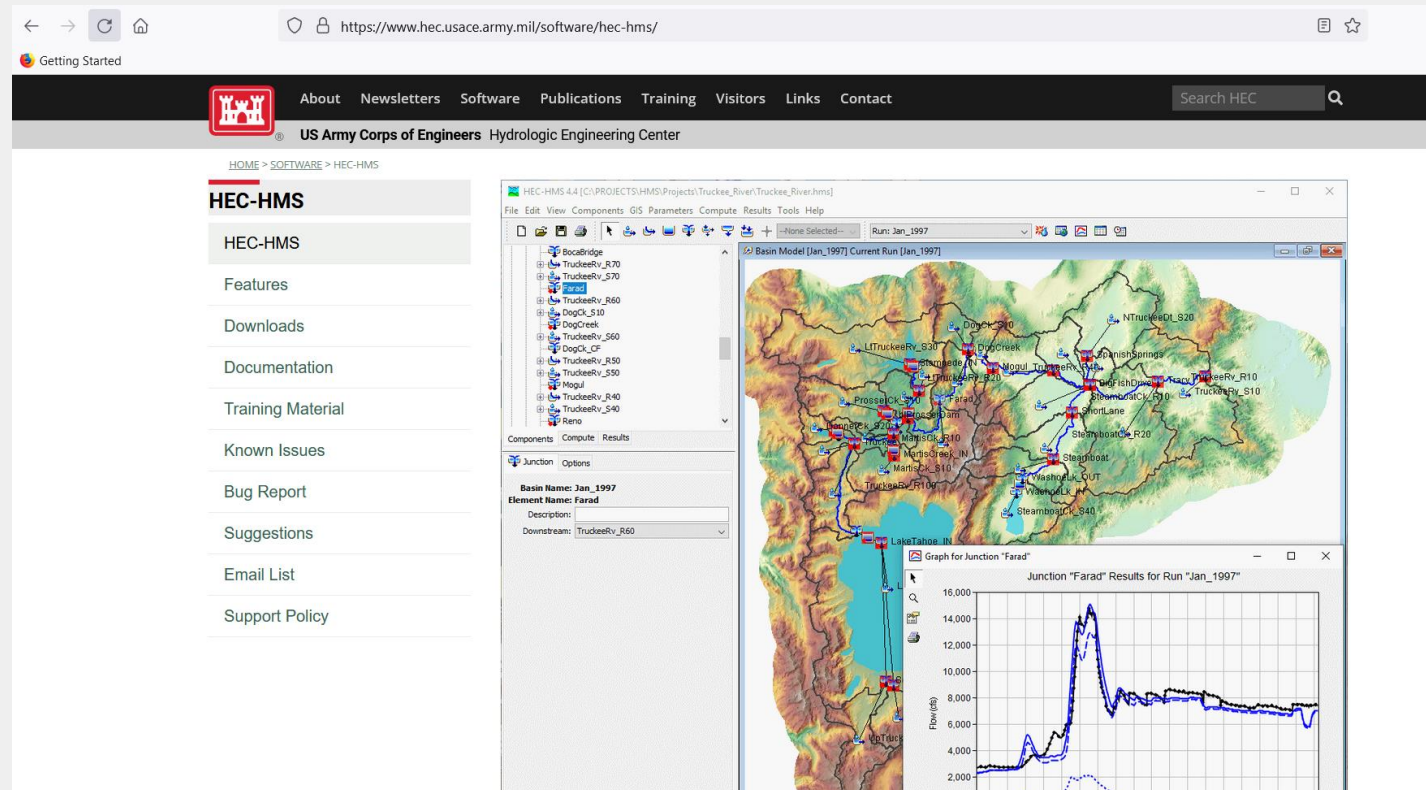
- Explain the different elements of HEC-HMS and the required input data
- Develop an event HEC-HMS model
- Apply HEC-HMS to produce a hydrograph and various summary statistics

HEC-HMS

HEC-HMS: Hydrologic Engineering Center – Hydrologic Modeling System

Available at: <http://www.hec.usace.army.mil/software/hec-hms/>

- Software
- Documentation



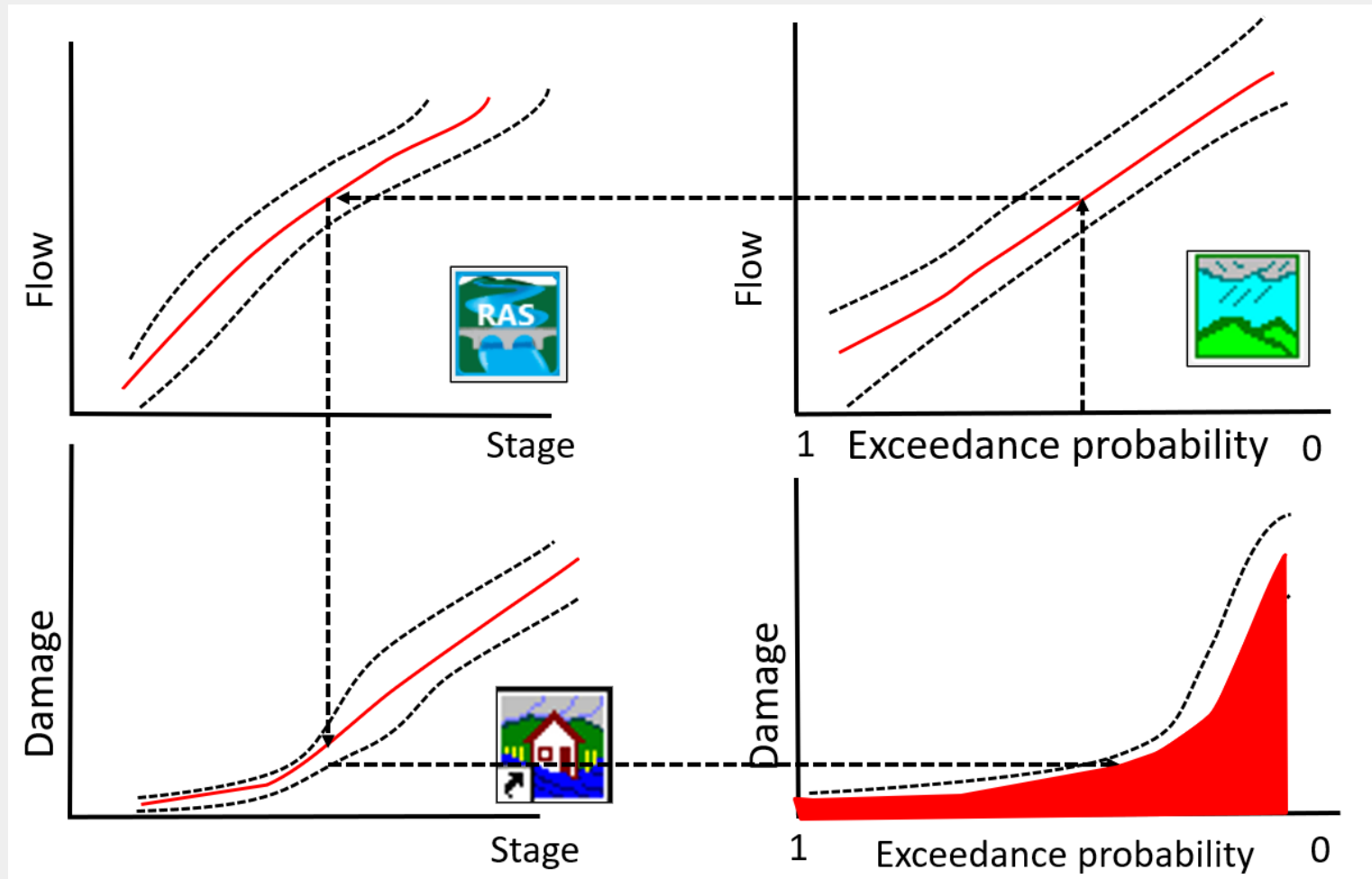
HEC-HMS

Potential applications of the

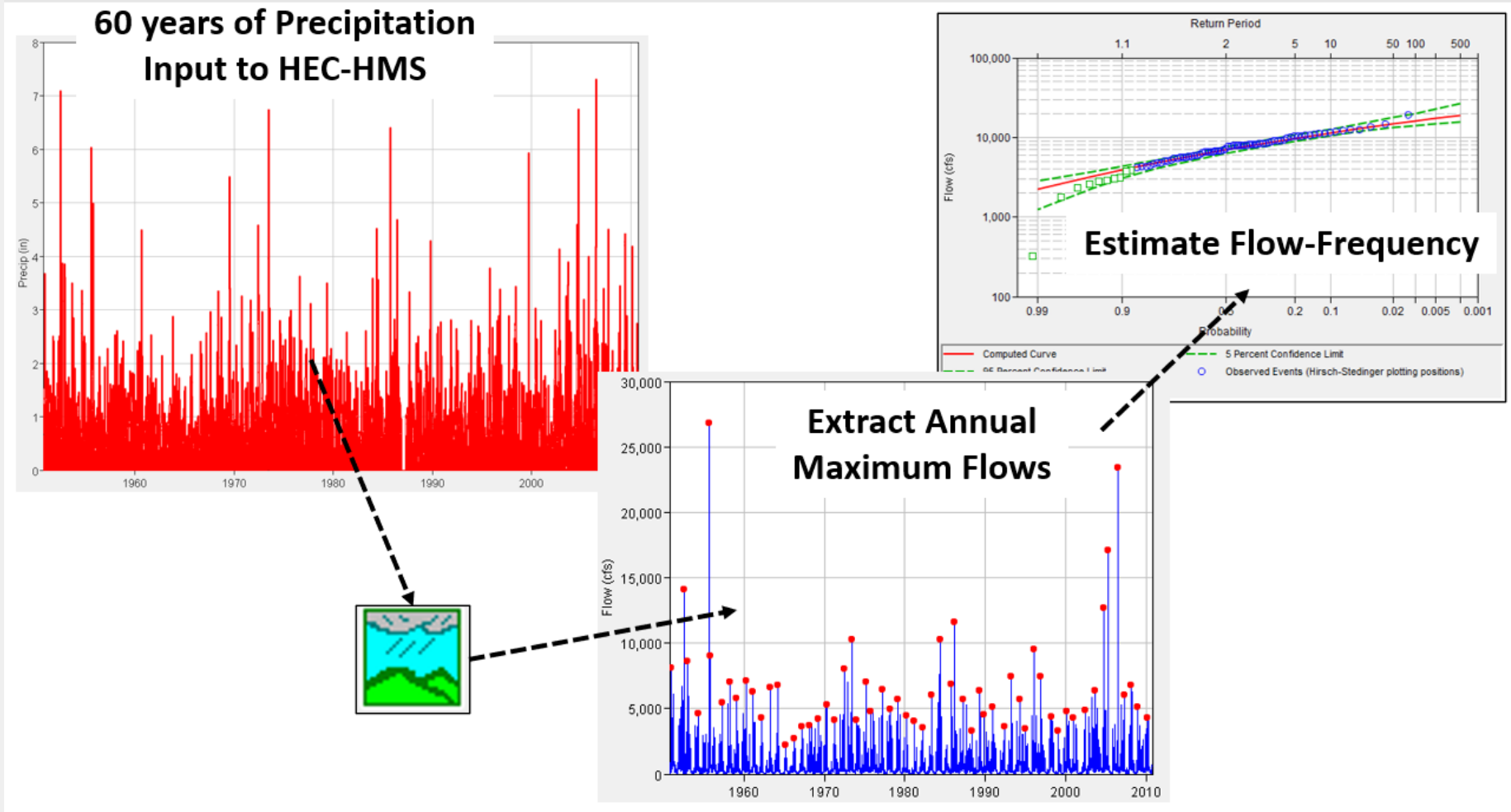
HEC-HMS: Hydrologic Engineering Center – Hydrologic Modeling System

- Flood and Drought Forecasting
- Floodplain and Flood Risk Management
- Hydrologic Hazard Curve Development for Dam and Stopbank Safety
- Development of Probable Maximum Flood or Inflow Design Flood for Dam Safety (incl. Flow Frequency Analysis)
- Determination of Pump Station / Interior Drainage Adequacy for Stopbank Safety
- Water Supply Management (e.g. Reservoir Storage Management)
- Environmental Restoration

Flood risk management



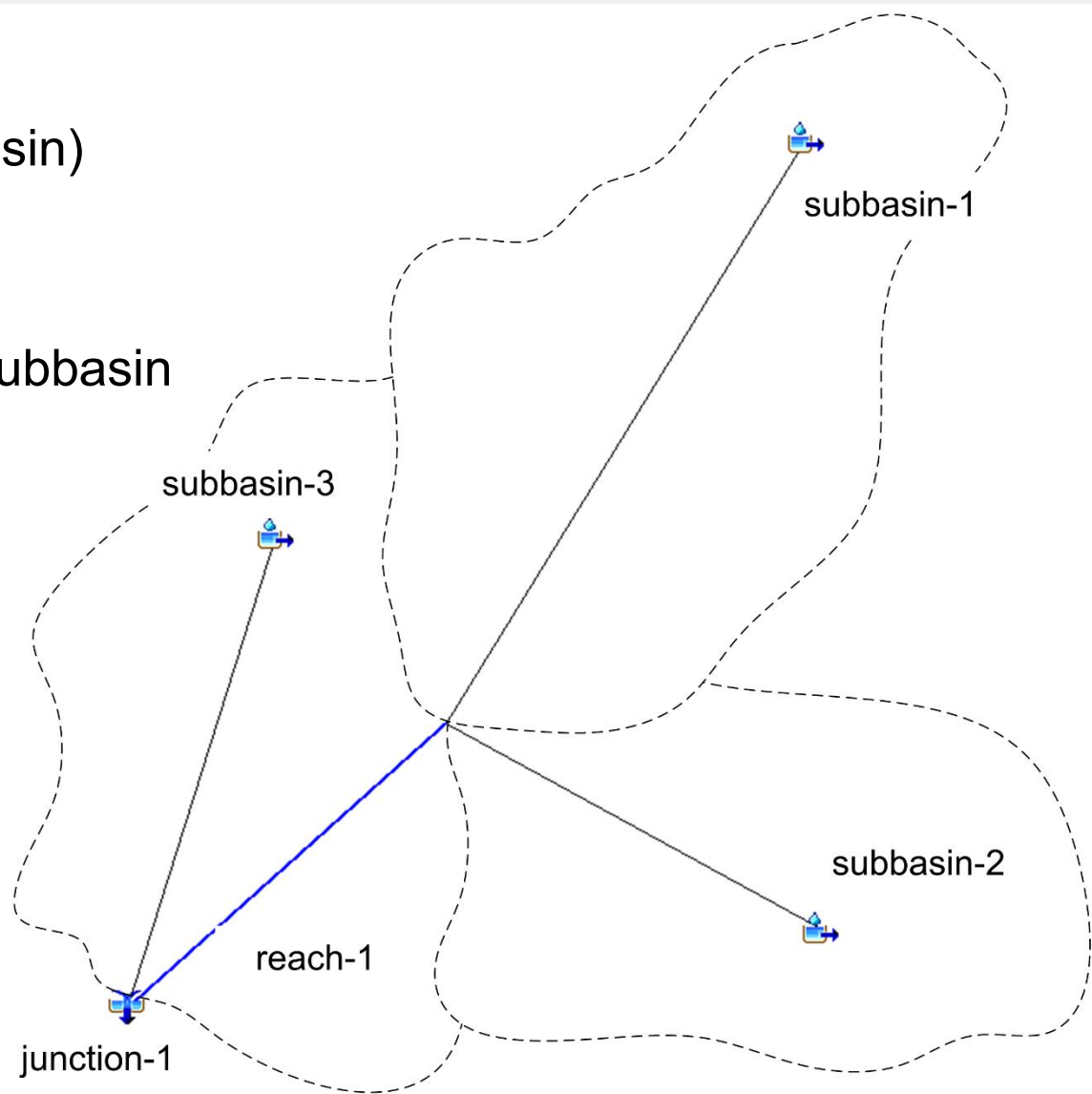
Flow frequency analysis based on continuous simulation



HEC-HMS

Catchment model:

- Subcatchments (subbasin)
 - River sections (reach)
 - junction to connect reach-reach or reach-subbasin
- furthermore: source
 sink
 reservoir
 diversion



HEC-HMS: empirical or physically based

SCS Curve Number Loss Method

Layered Green & Ampt Loss Method

Subbasin Loss Transform Baseflow Options

Basin Name: Alt Post Dam
Element Name: Deer Cr

Initial Abstraction (MM)

*Curve Number:

*Impervious (%)

Subbasin Canopy Surface Loss Transform Baseflow Options

Basin Name: Alt Post Dam
Element Name: Deer Cr

*Layer 1 Initial Content:

*Layer 2 Initial Content:

*Conductivity (MM/HR)

*Max Seepage (MM/HR)

*Max Percolation (MM/HR)

*Suction (MM)

*Dry Duration (HR)

*Impervious (%)

*Layer 1 Thickness (MM)

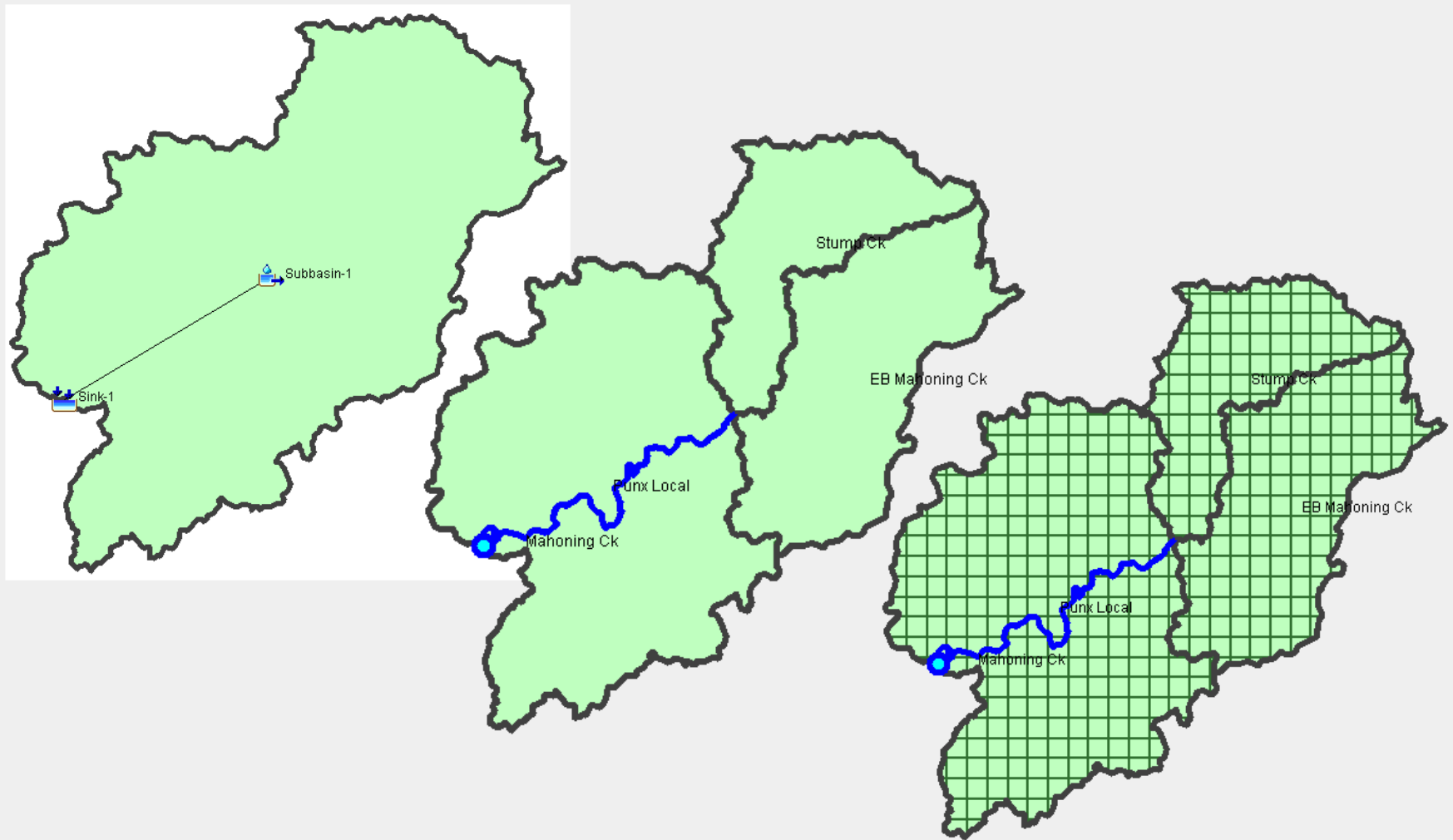
*Layer 1 Saturated Content:

*Layer 1 Field Capacity:

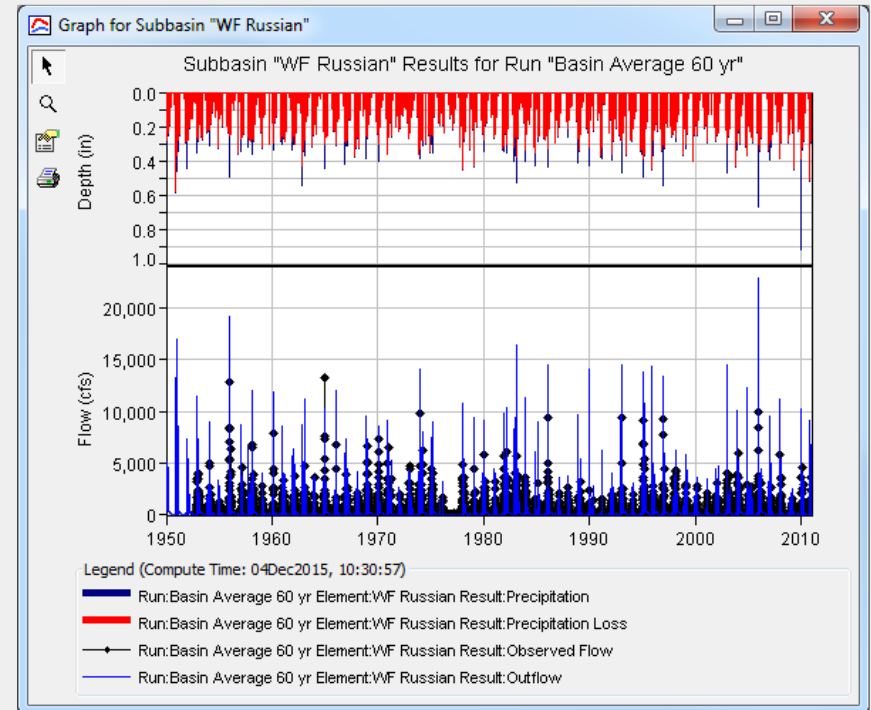
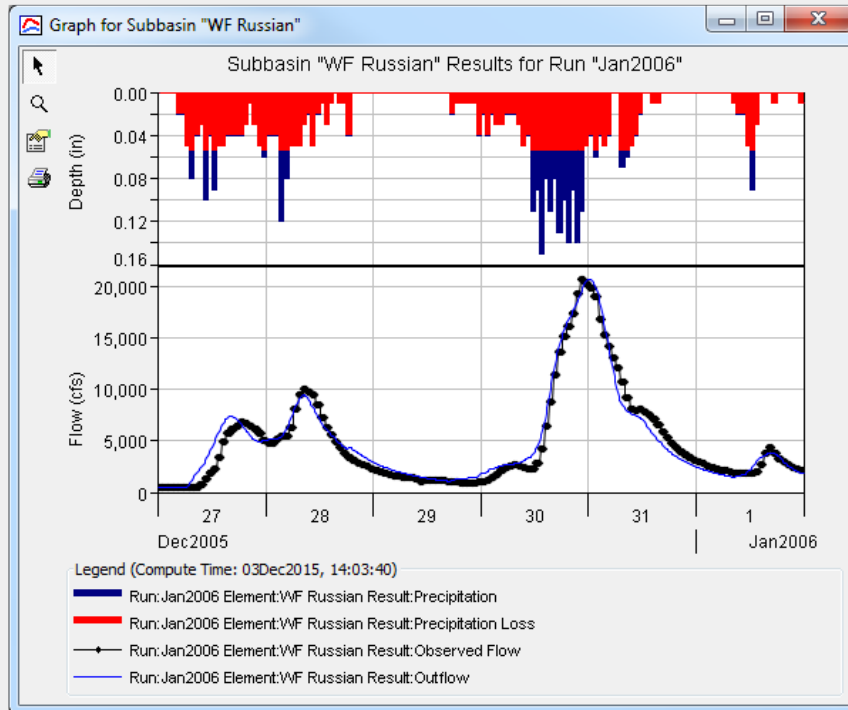
*Layer 1 Wilting Point:

*Layer 2 Thickness (MM)

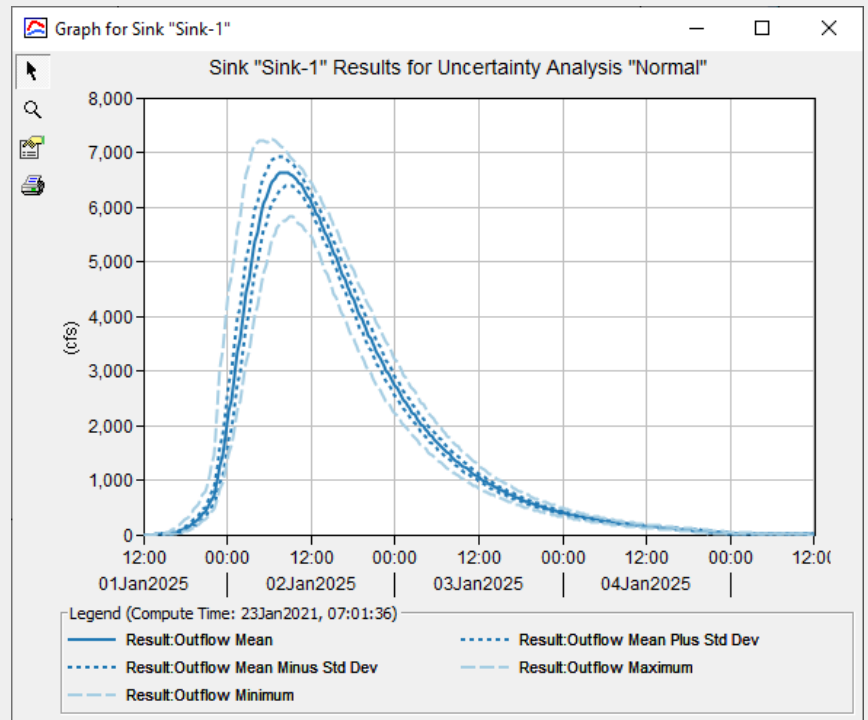
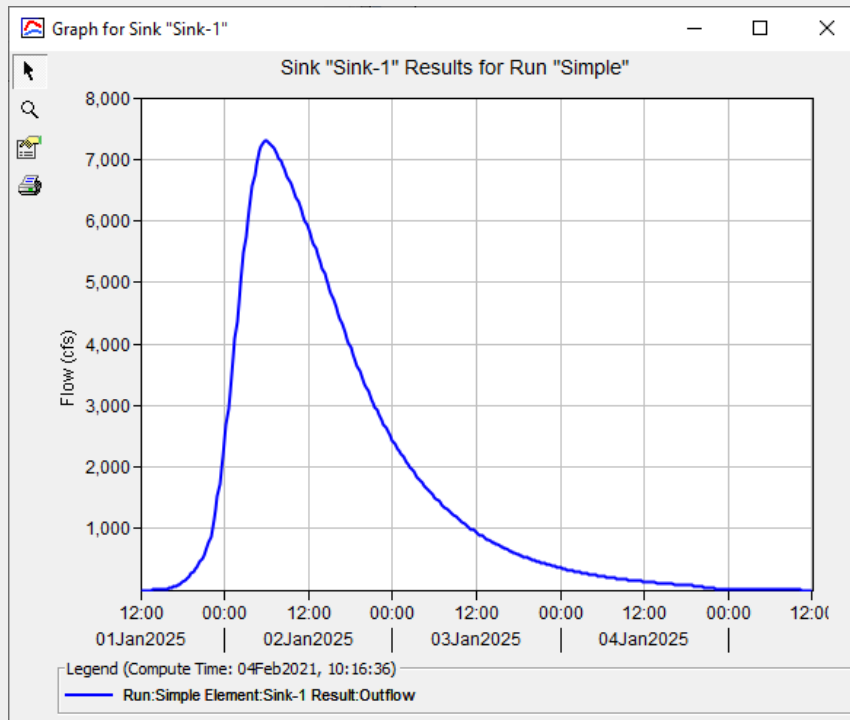
HEC-HMS: subbasin-averaged or gridded



HEC-HMS: event or continuous



HEC-HMS: deterministic or stochastic



HEC-HMS

Catchment model - Subbasin

Hydrologic Element	Calculation Type	Method
Subbasin	Canopy	Simple Canopy Gridded Simple Canopy
	Surface	Simple Surface Gridded Simple Surface
	Runoff-volume	Deficit and constant rate (DC)
		Exponential
		Green and Ampt
		Gridded DC
		Gridded SCS CN
		Gridded Green and Ampt
		Gridded SMA
		Initial and constant rate
		SCS curve number (CN)
		Smith Parlange
		Soil moisture accounting (SMA)
	Direct-runoff	Clark's UH
		Kinematic wave
		ModClark
		SCS UH
		Snyder's UH
		User-specified s-graph
		User-specified unit hydrograph (UH)
	Baseflow	Bounded recession
		Constant monthly
		Linear reservoir
		Nonlinear Boussinesq
		Recession

HEC-HMS

Catchment model – Reach

Hydrologic Element	Calculation Type	Method
Reach	Routing	Kinematic wave
		Lag
		Modified Puls
		Muskingum
		Muskingum-Cunge
		Straddle Stagger
	Loss/Gain	Constant
		Percolation

Meteorological Model:

- to determine the precipitation input for the sub-catchments
- snow calculations, snow melt
- consideration of evaporation

Table 3. Description of meteorologic model methods.

Precipitation Methods	Description
Frequency Storm	This method is used to develop a precipitation event where precipitation depths for various durations within the storm have a consistent exceedance probability.
Gage Weights	This method applies user specified weights to user defined precipitation gages.
Gridded Precipitation	This method allows the use of gridded precipitation products, such as RADAR.
Inverse Distance	This method calculates subbasin average precipitation by applying an inverse distance squared weighting to user defined precipitation gages.
SCS Storm	This method applies a user specified SCS time distribution to a 24-hour total storm depth.
Specified Hyetograph	This method applies a user defined hyetograph to a specified subbasin element.
Standard Project Storm	This method applies a time distribution to an index precipitation depth.

HEC-HMS

Control components:

- Control settings for the simulation runs:
 - * start and end time
 - * computation time step

HEC-HMS

Options

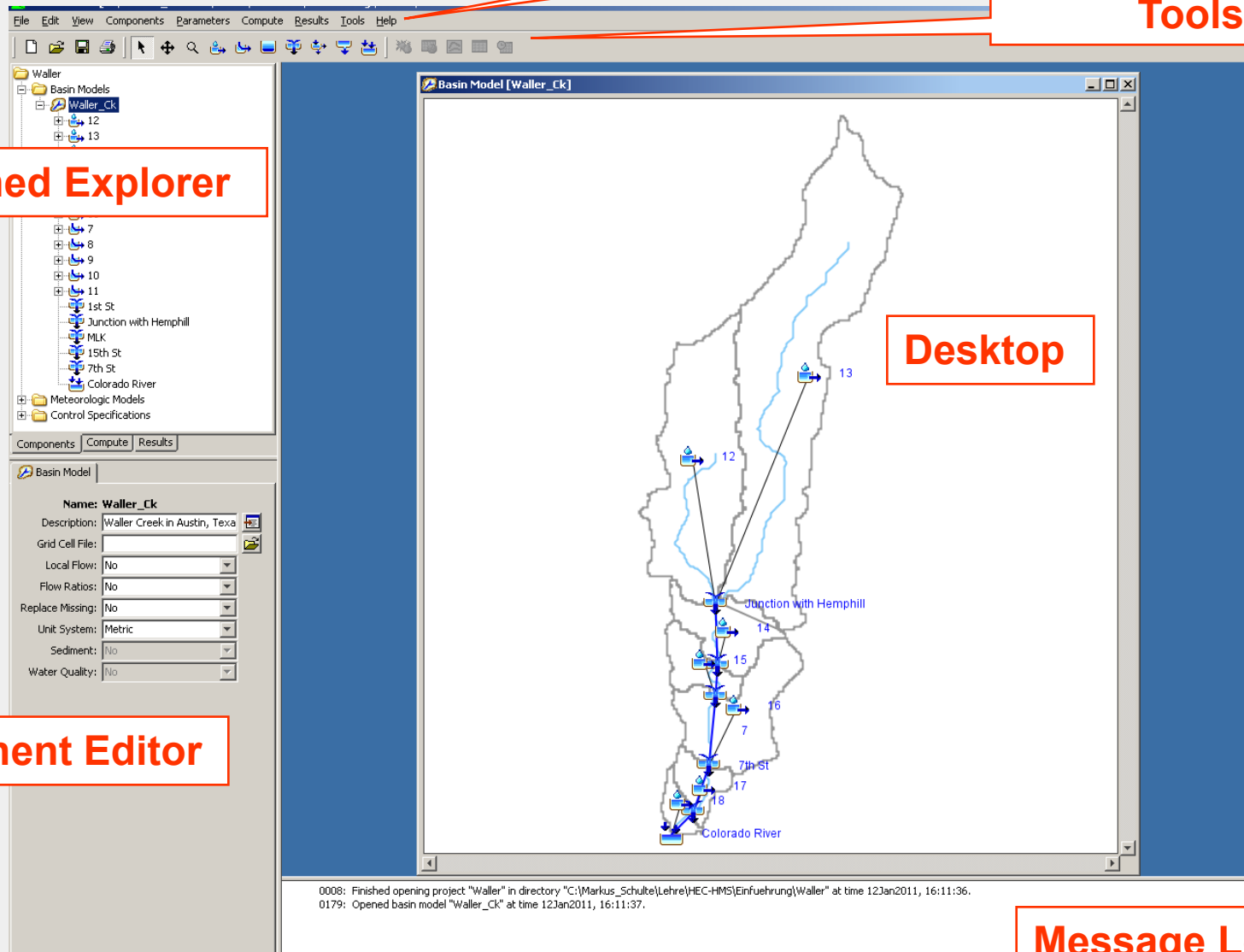
Tools

Watershed Explorer

Desktop

Component Editor

Message Log



HEC-HMS: Building a basin (catchment) model

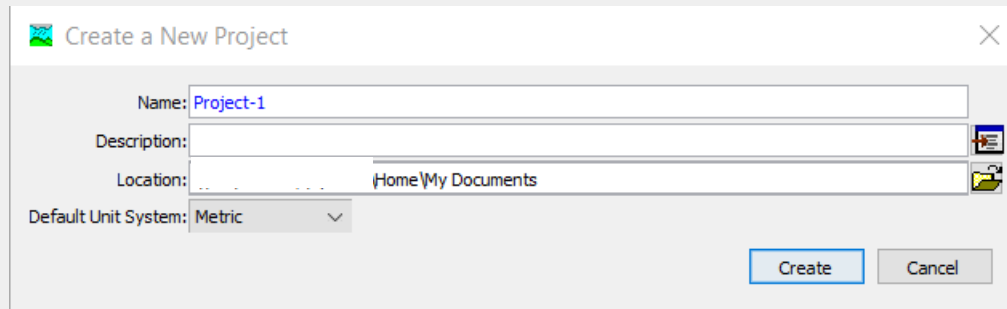
Steps

1. Create a new Project
2. Create a new Basin model
3. Add Hydrologic Elements to the Basin model
4. Select Modeling Methods
5. Parametrise Modeling Methods

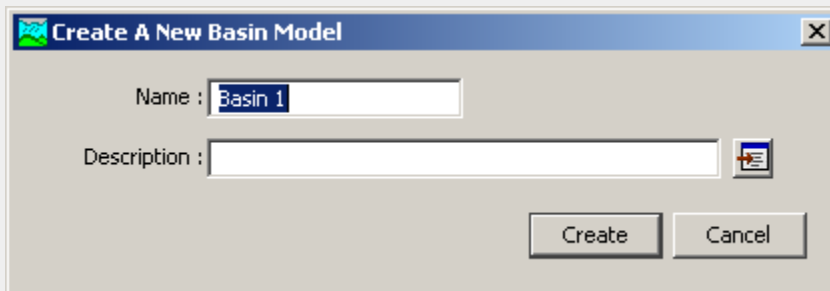
HEC-HMS

First project:

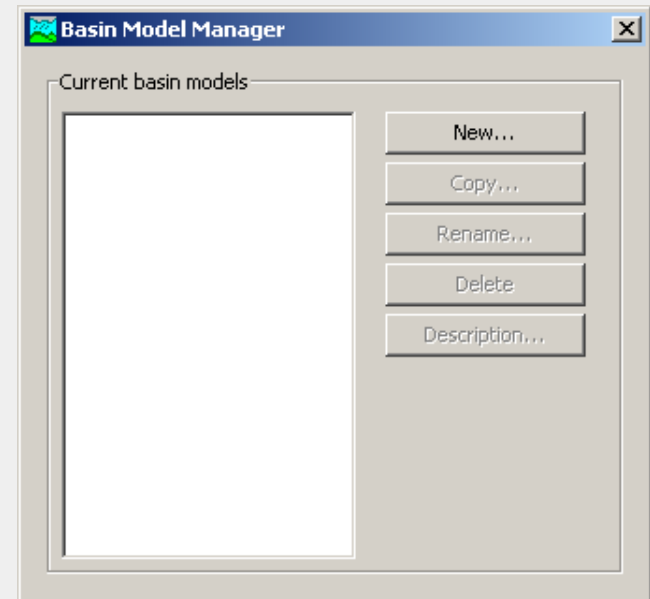
- File → New...



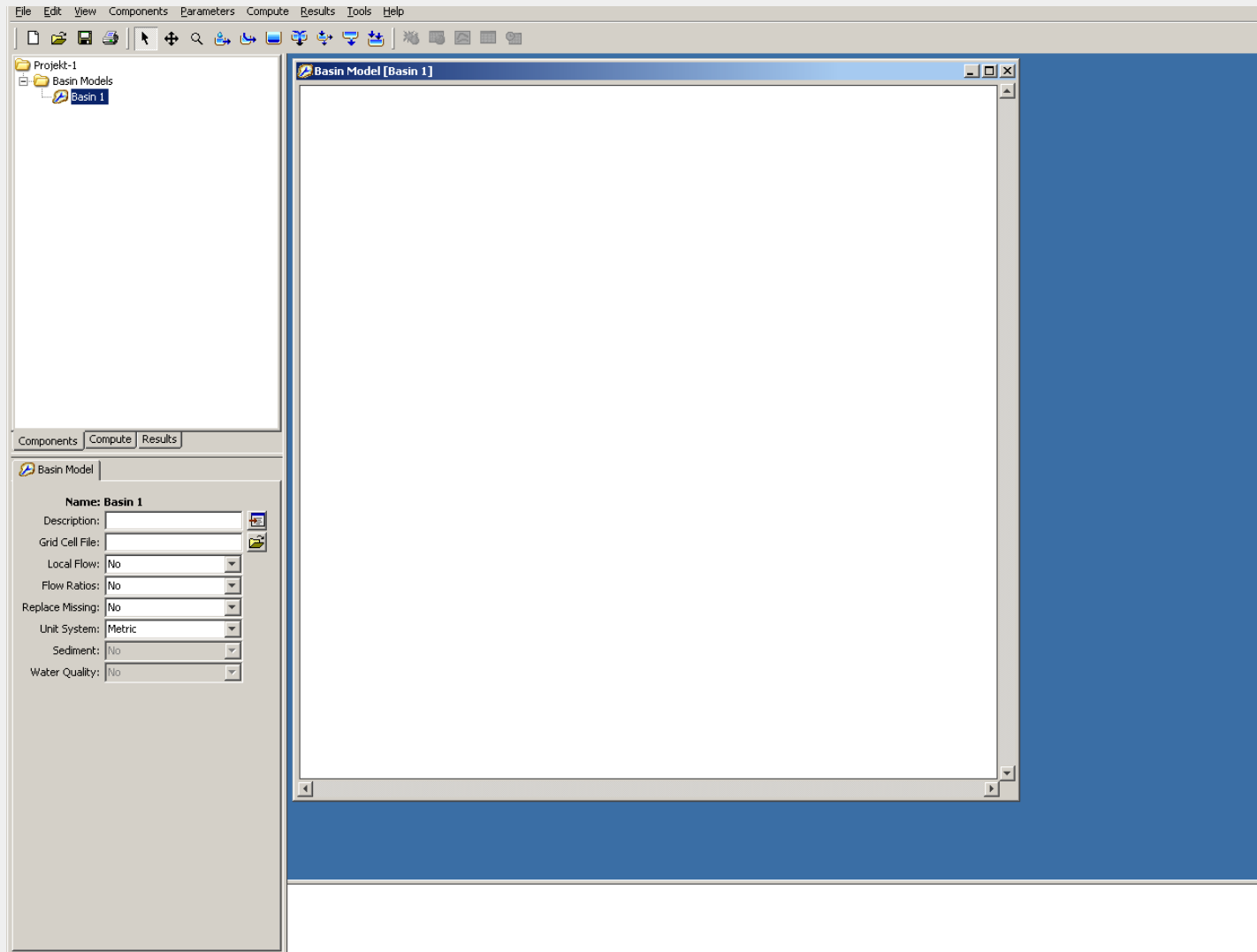
- Components → Basin Model Manager
→ New.. → Create



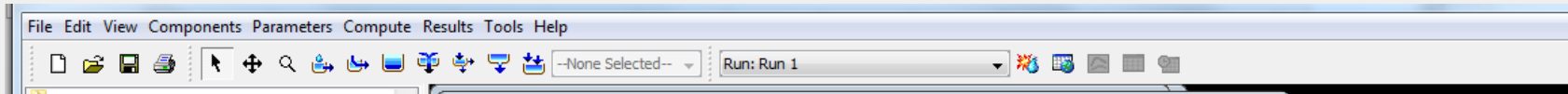
- Close Basin Model Manager



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HEC-HMS



Subbasin – Used for rainfall-runoff computation on a watershed.



Reach – Used to convey (route) streamflow downstream in the basin model.

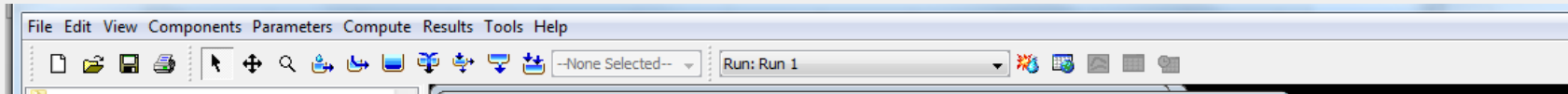


Reservoir – Used to model the detention and attenuation of a hydrograph caused by a reservoir or detention pond.



Junction – Used to combine flows from upstream reaches and sub-basins.

HEC-HMS



Diversion – Used to model abstraction of flow from the main channel.



Source – Used to introduce flow into the basin model (from a stream crossing the boundary of the modeled region). Source has no inflow.



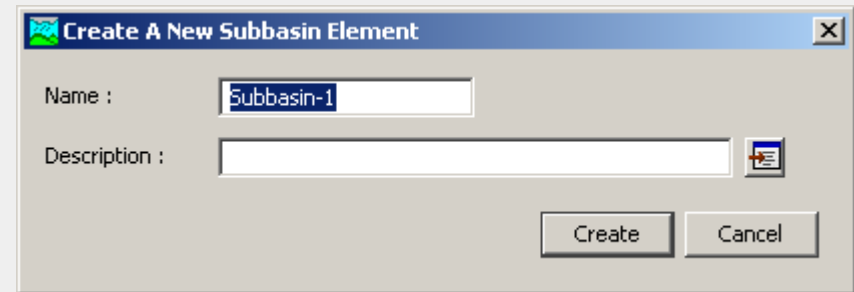
Sink – Used to represent the outlet of the physical watershed. Sink has no outflow.

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- Subbasin Creation Tool



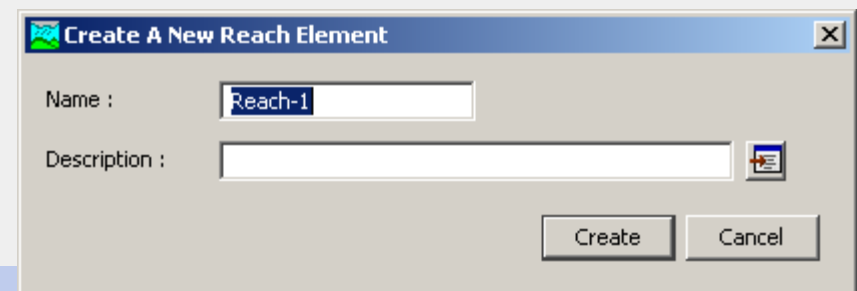
- left click in desktop window
- Create



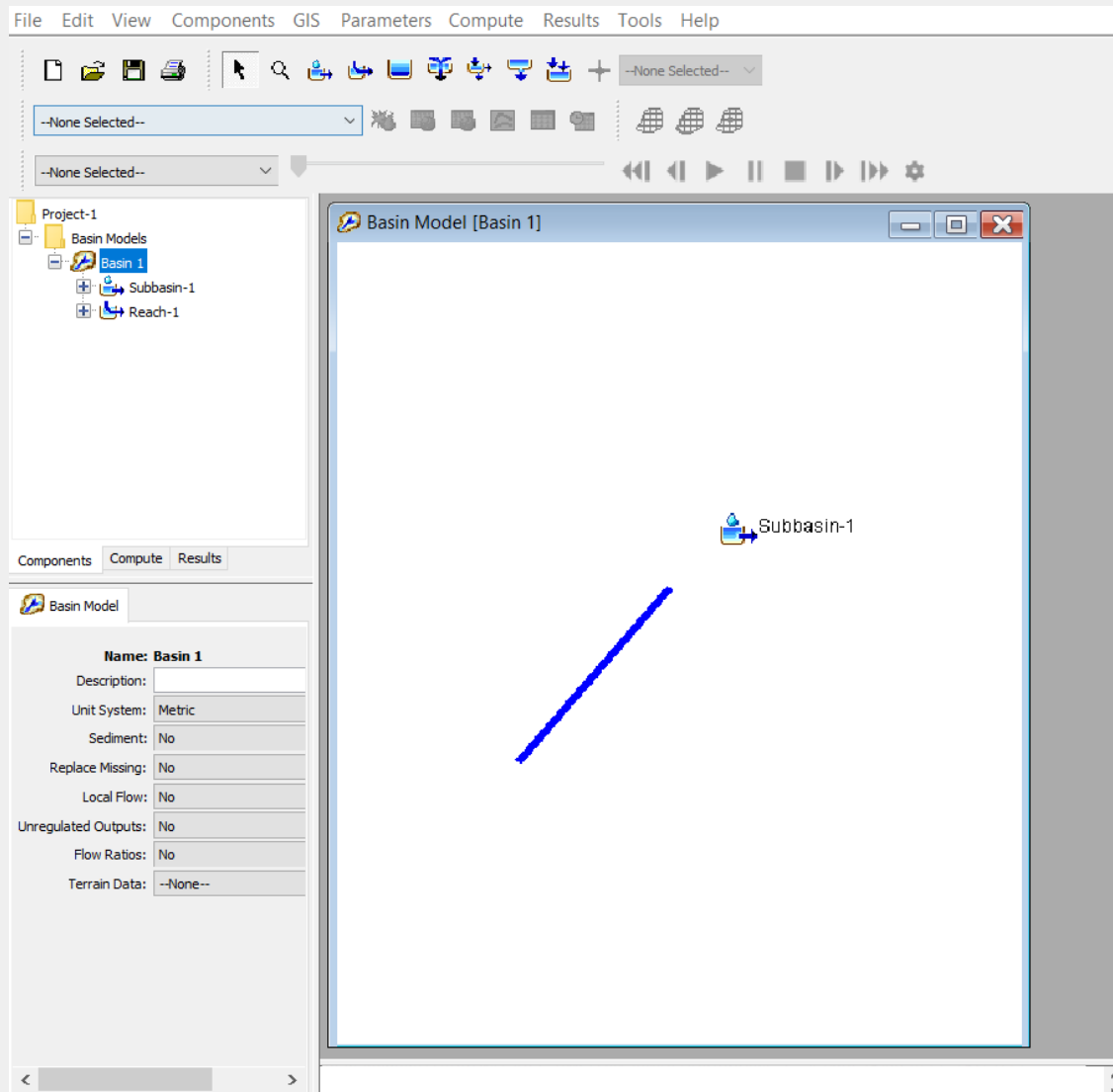
- Reach Creation Tool



- draw in dektop window (first click upstream!)
- Create

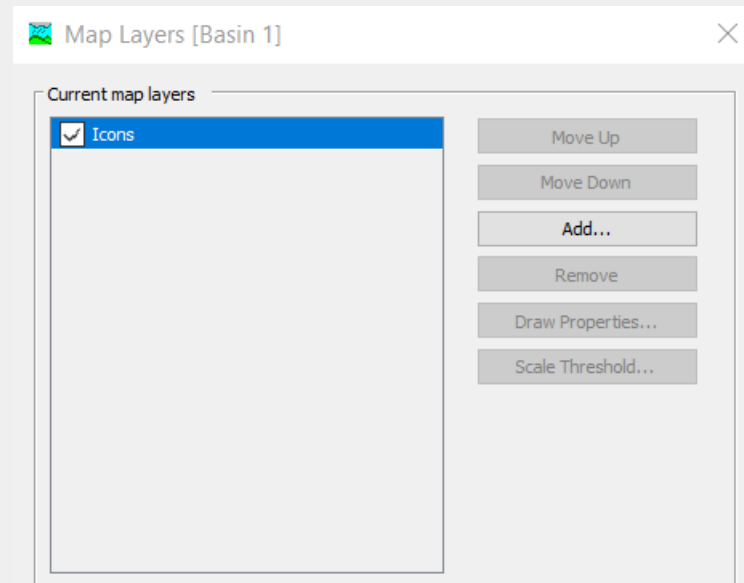


HEC-HMS



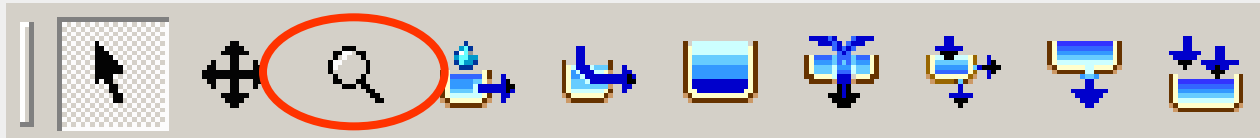
HEC-HMS

- Adding a background map (just a visual element – not done here)
- View → Map Layers... → Add...
- select map
- close Background Maps window



HEC-HMS

- Zoom function



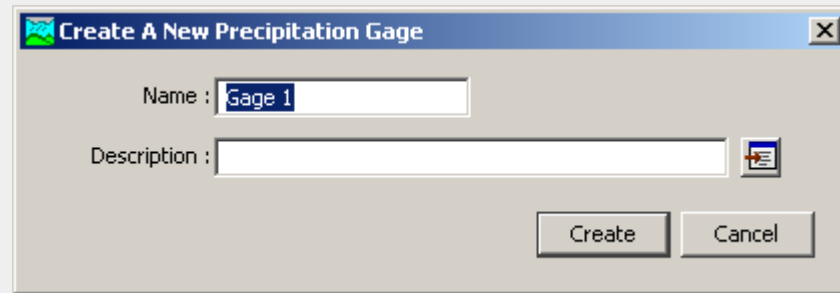
HEC-HMS: Creating a meteorological model

Steps

1. Create new Precipitation Gauges
2. Parameterise gages
3. Create a new Meteorological Model
4. Select Modelling Methods
5. Parameterise Modelling Methods

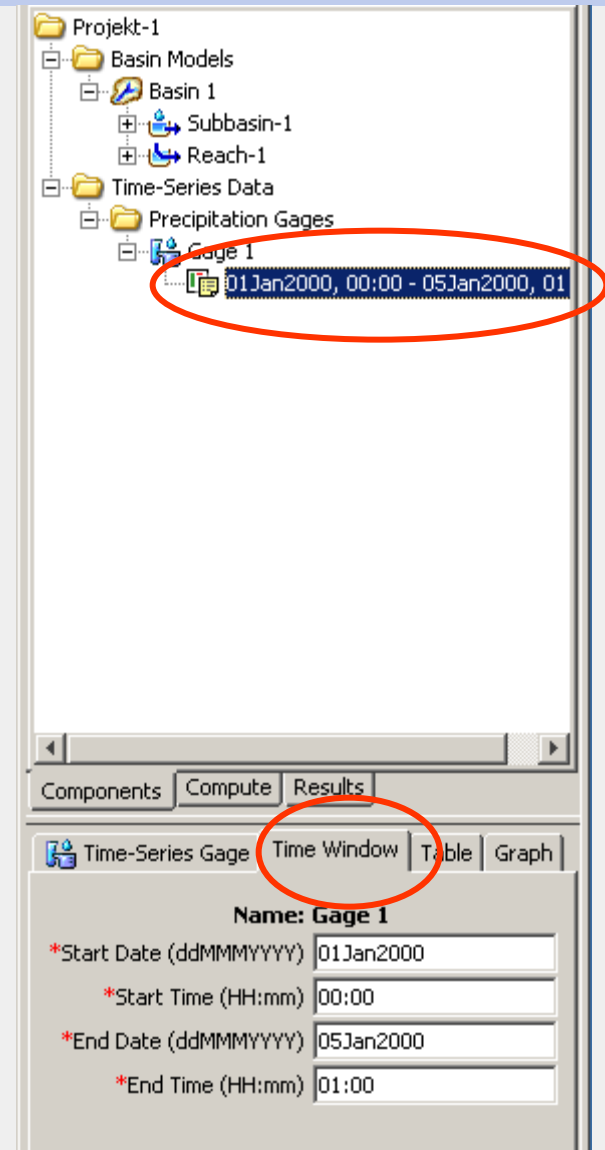
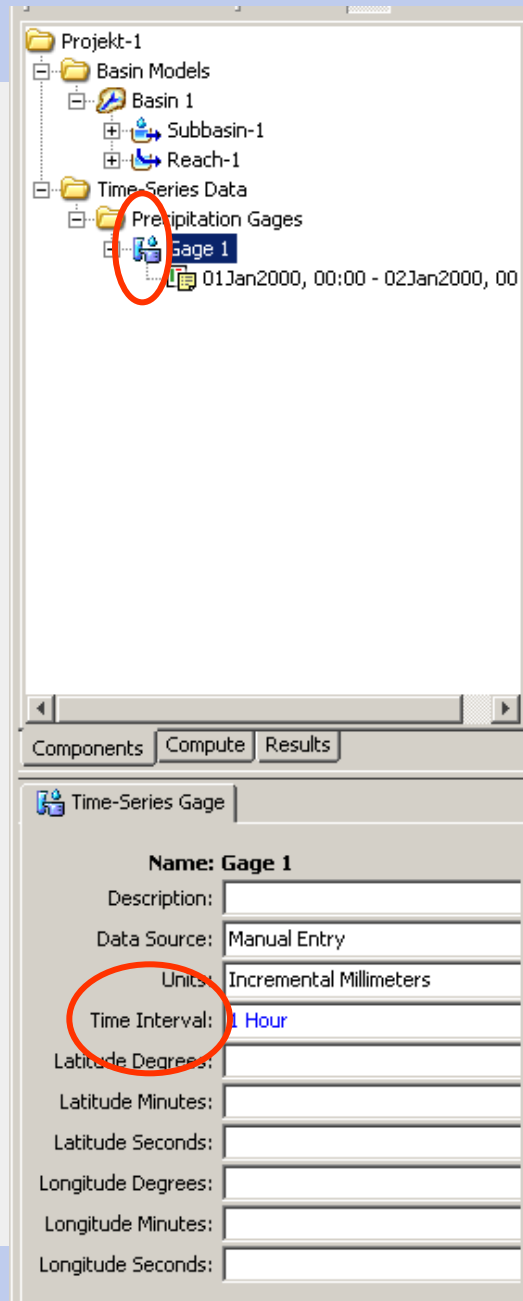
HEC-HMS

- Components → Time-Series Data Manager
- Data Type: Precipitation Gages → New... → Create



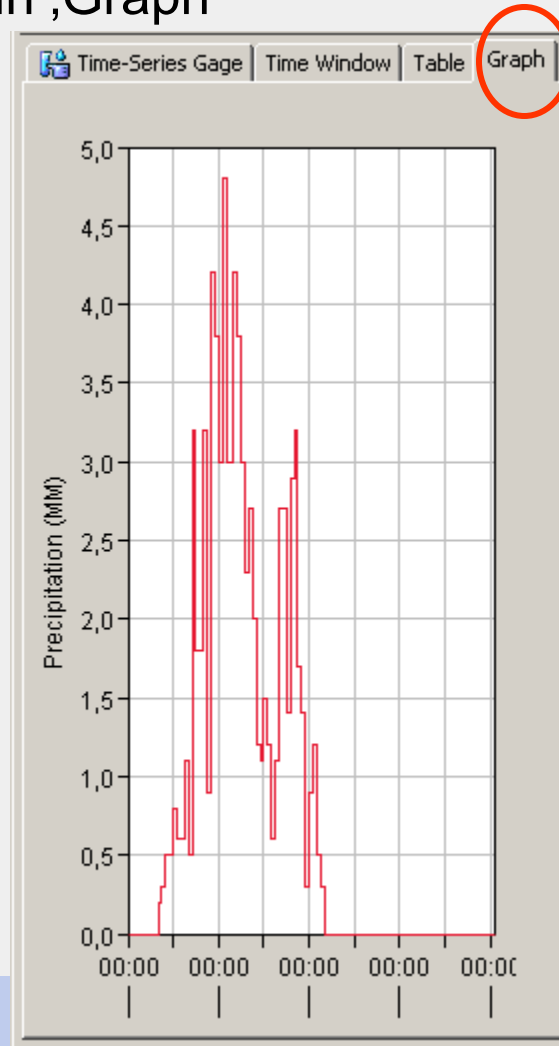
HEC-HMS

- Open rainfall data file 'Start-Project.xlsx'
- What is the time step of the rainfall data? Enter into 'Time-Series Gage'
- What is the start and end date of the rainfall time series? Enter into 'Time Window'.



HEC-HMS

- Copy-paste precipitation data from Start-Project.xlsx into ,Table‘
- View the time series in ,Graph‘

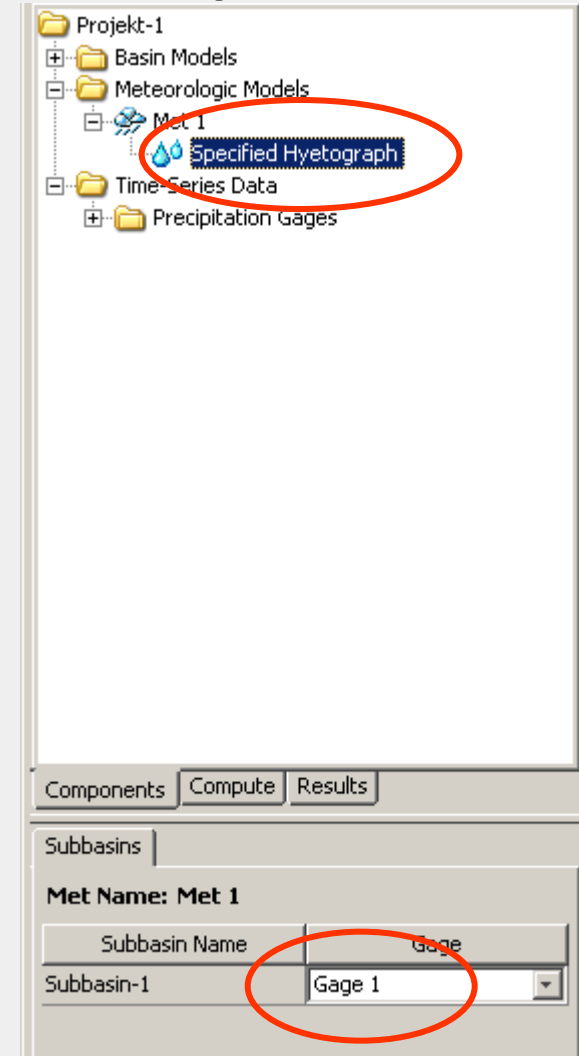
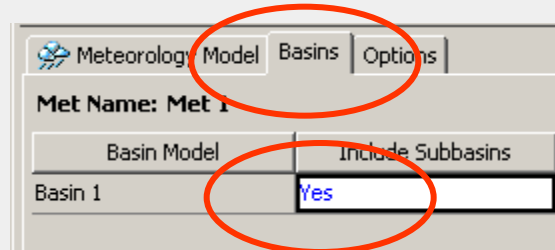
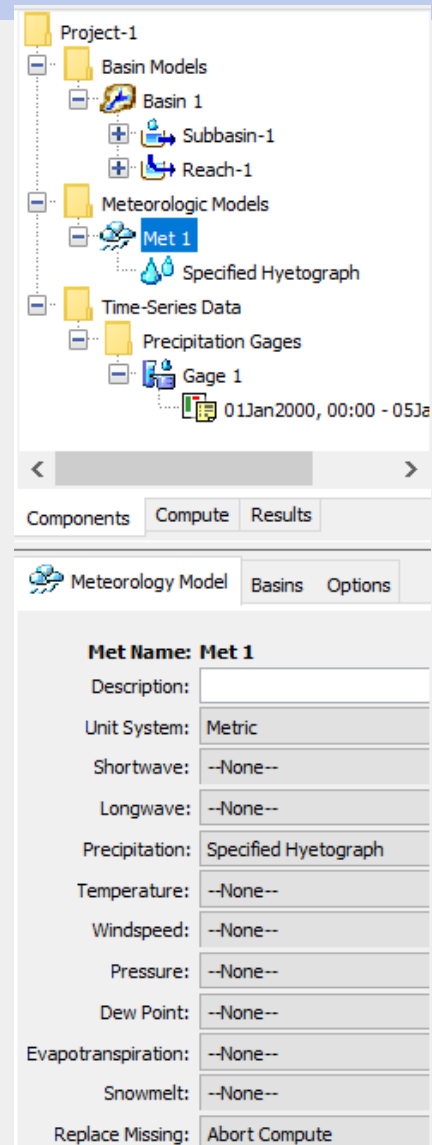


The screenshot shows the HEC-HMS project tree and the 'Time-Series Gage' window. The project tree on the left shows the hierarchy: Projekt-1 > Basin Models > Basin 1 > Subbasin-1 > Reach-1 > Time-Series Data > Precipitation Gages > Gage 1. The 'Time-Series Gage' window on the right shows the 'Table' tab with a table of precipitation data. The 'Table' tab is circled in red.

Time (ddMMYYYY, H...	Precipitation (MM)
01Jan2000, 00:00	
01Jan2000, 01:00	0.0
01Jan2000, 02:00	0.0
01Jan2000, 03:00	0.0
01Jan2000, 04:00	0.0
01Jan2000, 05:00	0.0
01Jan2000, 06:00	0.0
01Jan2000, 07:00	0.0
01Jan2000, 08:00	0.0
01Jan2000, 09:00	0.2
01Jan2000, 10:00	0.3

HEC-HMS

- Components → Meteorologic Model Manager
→ New... → Create



HEC-HMS: Selecting methods

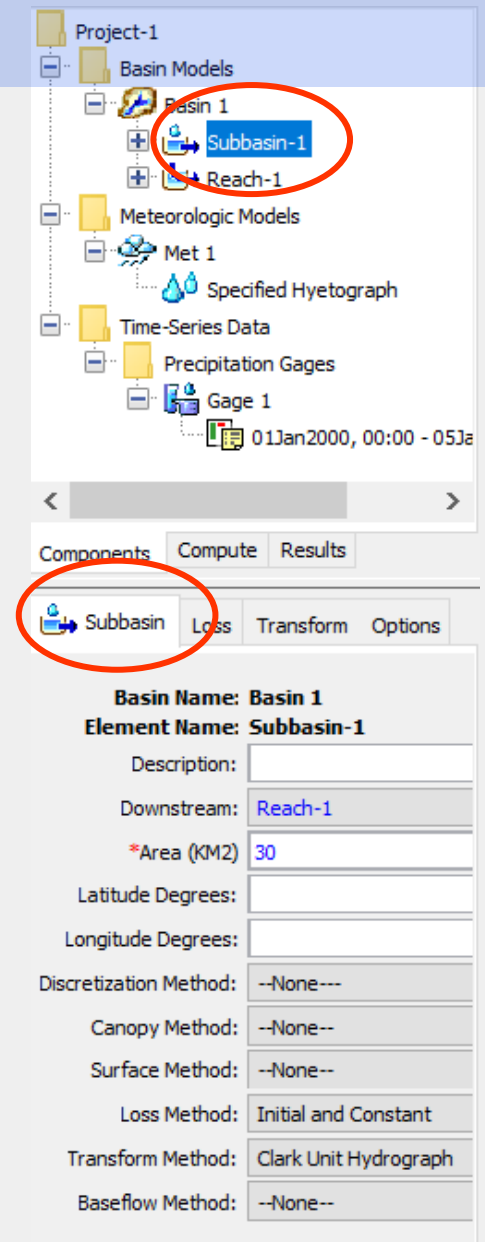
Steps

1. Loss Methods
2. Transform Methods
3. Baseflow Methods
4. Channel Routing Methods
5. Other

HEC-HMS

- Settings for subcatchment:

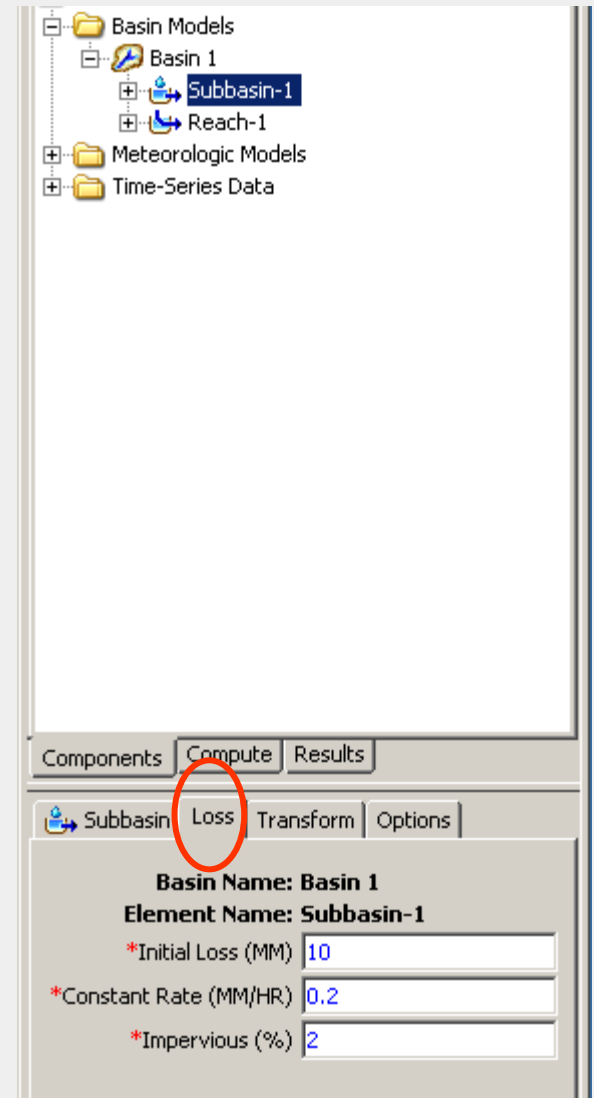
- * Choose Downstream: Reach-1
- * Area [km²]: 30 km²
- * Loss Method: Initial and Constant
- * Transform Method: Clark Unit Hydrograph



HEC-HMS

- Parameters for loss method:

- * Initial loss: 10 mm
- * Infiltration rate: 0.2 mm/h
- * Percentage impervious area: 2 %



Synthetic Unit Hydrographs (UHs)

- Synthetic unit hydrograph methods derive the unit hydrograph for a basin using theoretical or empirical formulas to relate hydrograph peak flow and timing to watershed characteristics.
- Utilizes instantaneous unit hydrographs
- Synthetic unit hydrograph methods in HMS:
 - Clark (ModClark)
 - SCS
 - Snyder

Transform method Clark UH

- Clark's model derives a watershed UH by representing two critical processes:
 - Translation or movement of excess runoff from its origin throughout the drainage to the watershed outlet.
 - Modelled using a time-area relationship
 - Time of concentration, “T_c”
 - Attenuation of the magnitude of the discharge as the excess is stored throughout the watershed
 - Modeled using a Linear Reservoir
 - Storage Coefficient, “R”

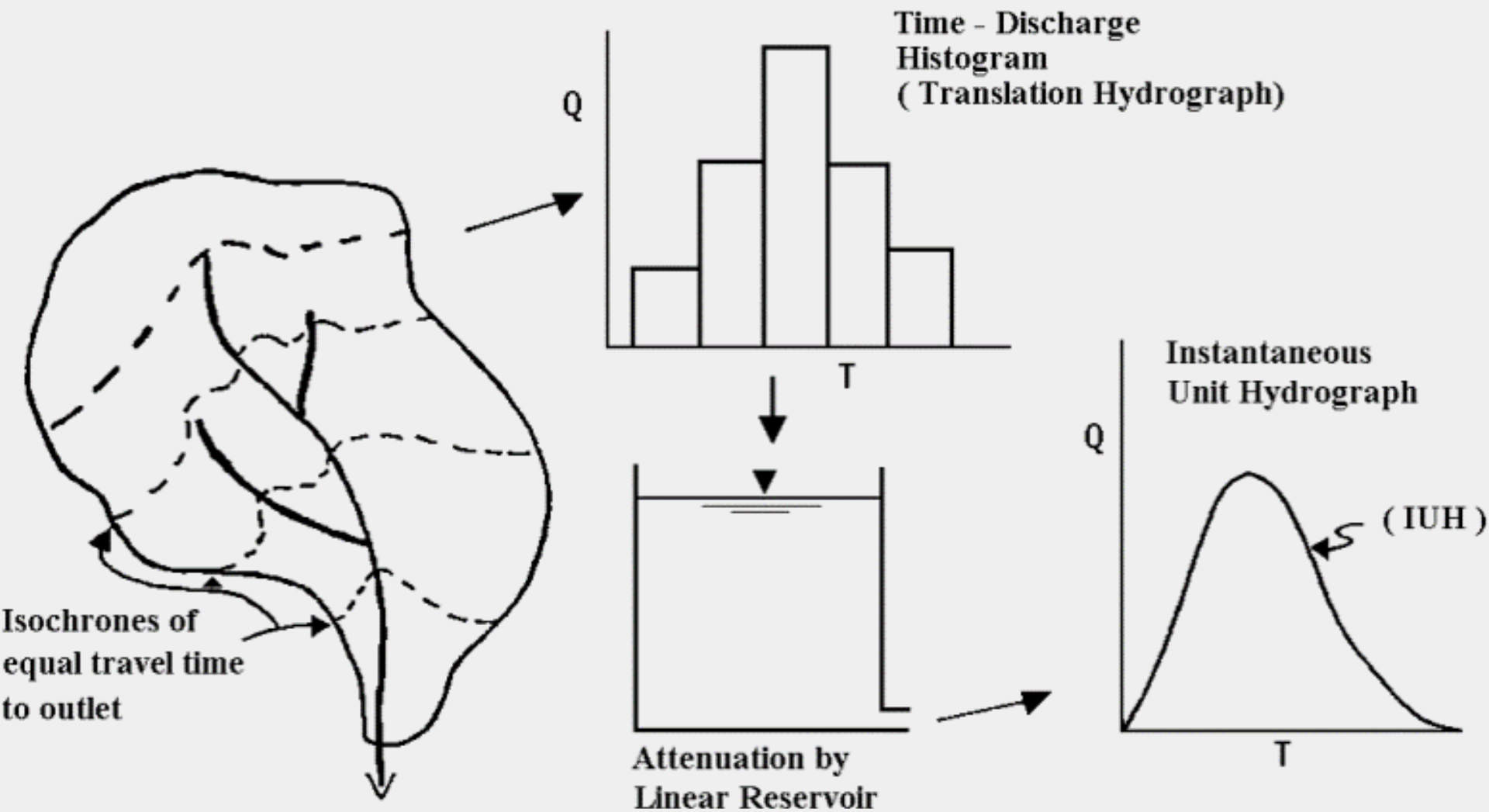
See also:

USACE HEC-HMS course video: <https://www.youtube.com/watch?v=uReJjcwtyWc>
and

USACE HEC-HMS technical reference manual:

<https://www.hec.usace.army.mil/confluence/hmsdocs/hmstrm/transform/clark-unit-hydrograph-model>

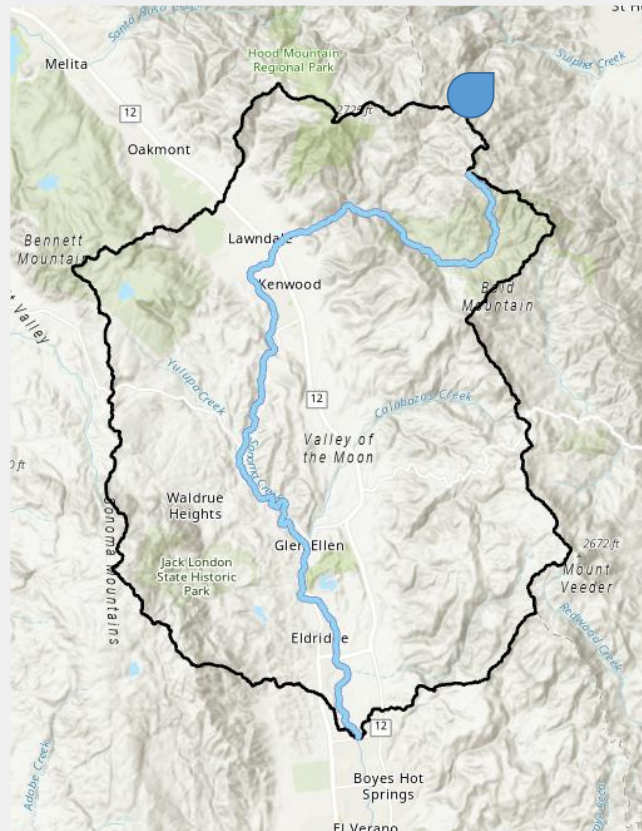
Transform method Clark UH



USACE (1994)

Transform method Clark UH Time of Concentration

- Time of Concentration – time required for runoff to travel from the hydraulically most distant point in the watershed to the outlet



Transform method Clark UH Time of Concentration

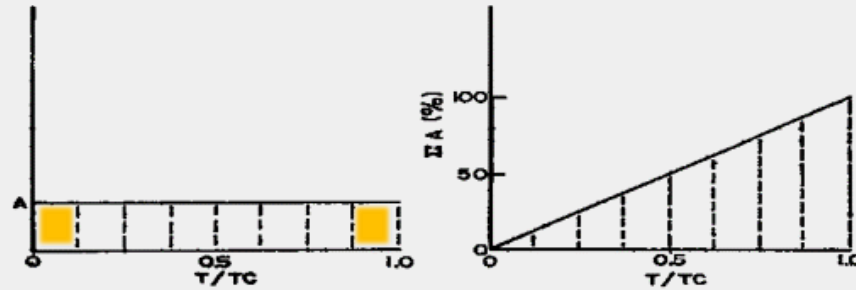
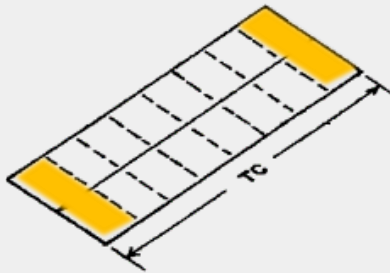


Fig.3 Rectangular Shape Watershed, $n=1$

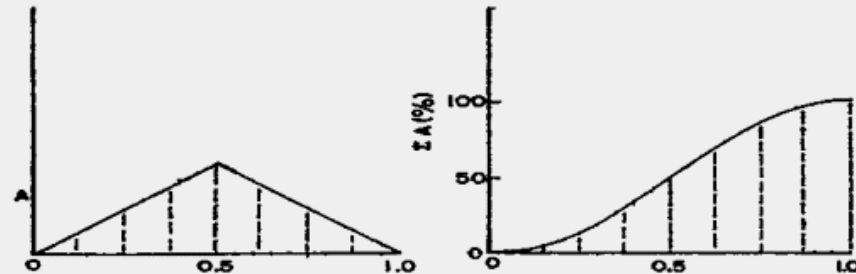
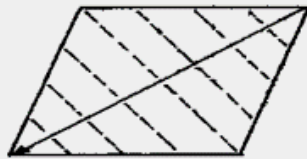


Fig.4 Diamond Shape Watershed, $n=2$

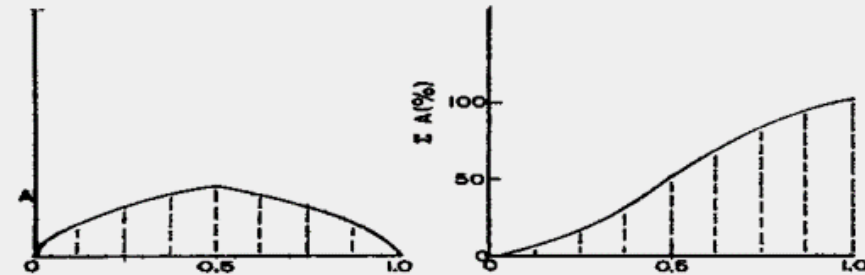
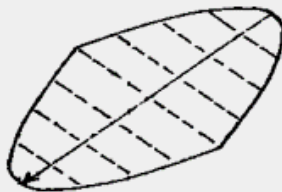


Fig.5 Generalized Shape Watershed, $n=1.5$

Transform method Clark UH Linear Reservoir

Linear Reservoir

$$S_t = RO_t$$

Storage Coefficient

Continuity Equation

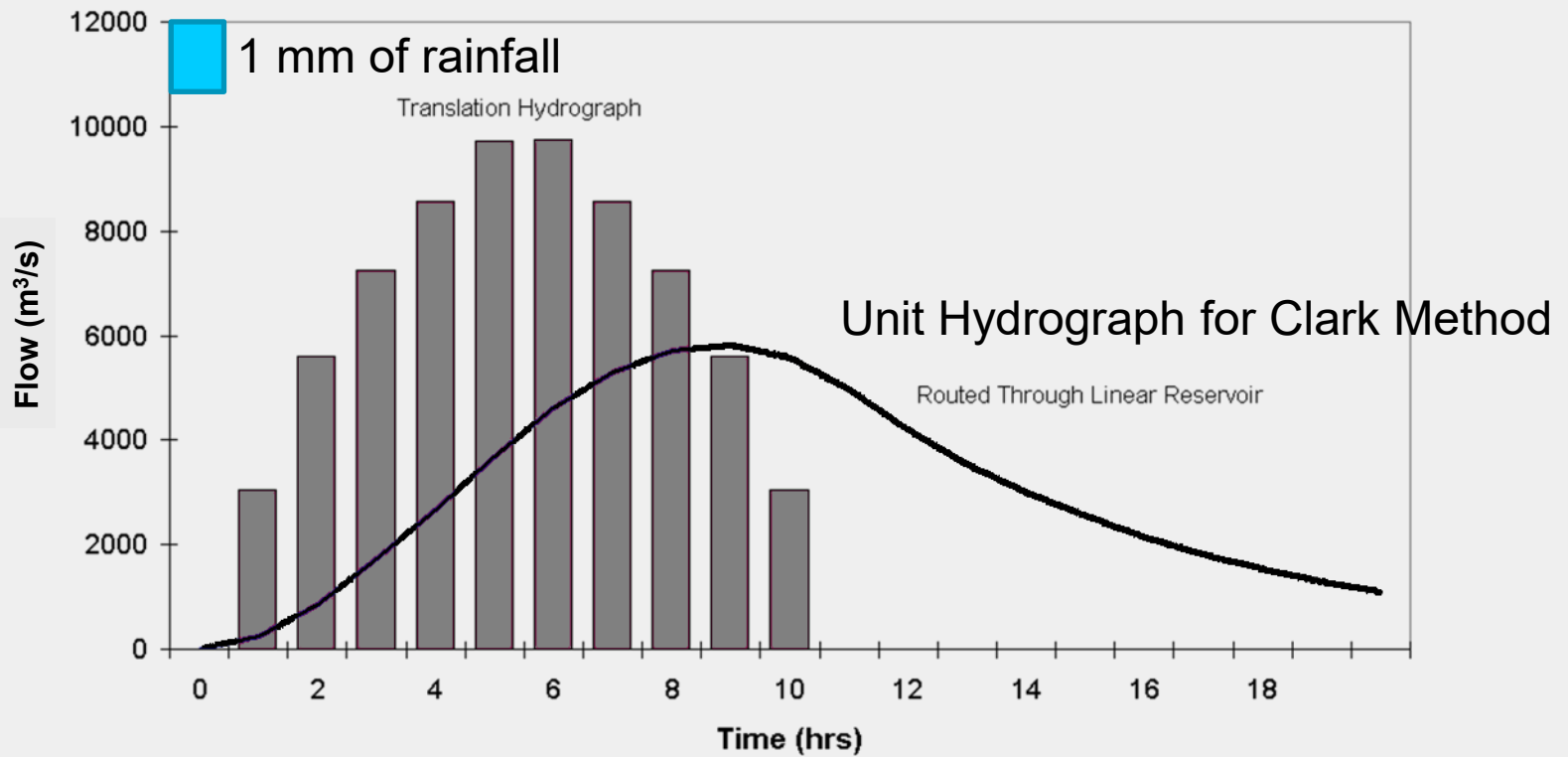
$$\frac{dS}{dt} = I_t - O_t$$

$$O_t = C_A I_t + C_B O_{t-1}$$

$$C_A = \frac{\Delta t}{R + 0.5\Delta t}$$
$$C_B = 1 - C_A$$

$$\bar{O}_t = \frac{O_{t-1} + O_t}{2}$$

Transform method Clark UH

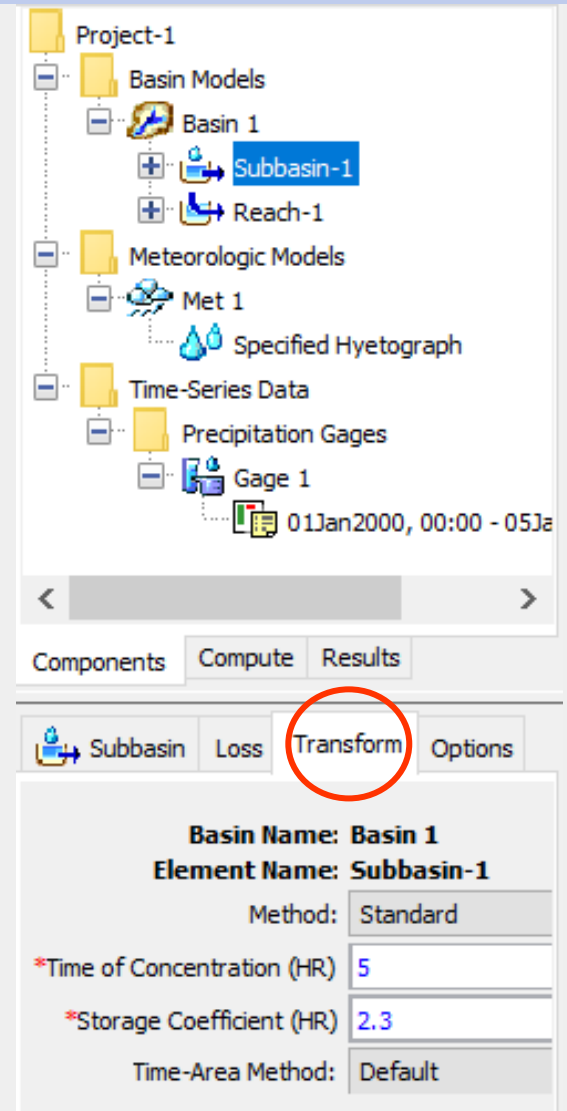


HEC-HMS

- Parameters for transform method:

* Time of concentration: 5 h

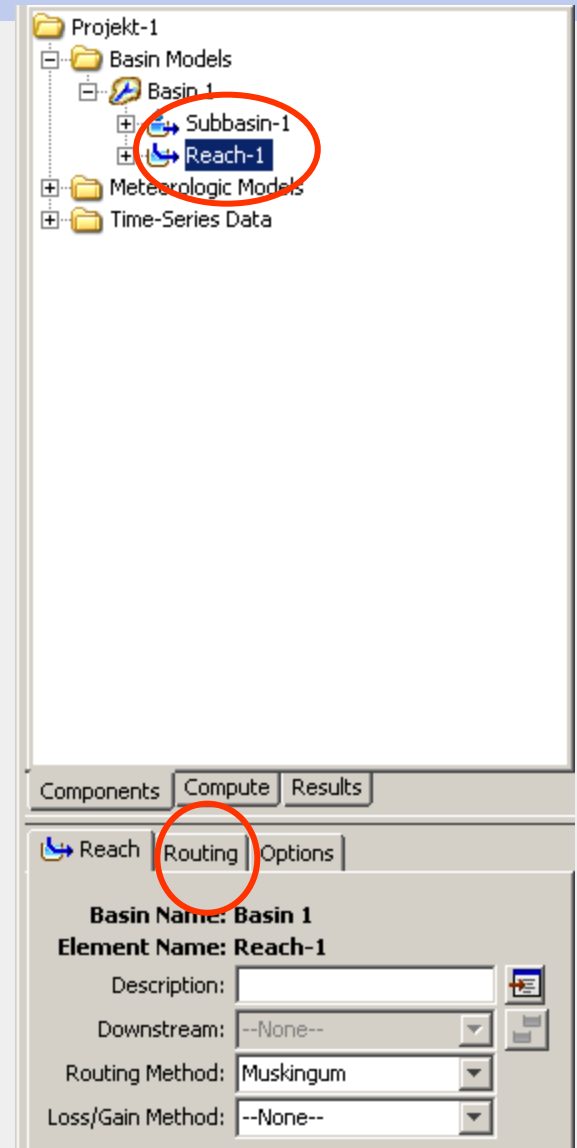
* Storage coefficient: 2.3 h



HEC-HMS

- Settings for the reach:

* Routing method: Muskingum



HEC-HMS

- Parameters for routing:

- * Parameter K: 2.7 h
- * Parameter X: 0.1

The screenshot displays the HEC-HMS software interface. The top panel shows a project tree for 'Project-1' with the following structure:

- Project-1
 - Basin Models
 - Basin 1
 - Subbasin-1
 - Reach-1 (highlighted in blue)
 - Meteorologic Models
 - Met 1
 - Specified Hyetograph
 - Time-Series Data
 - Precipitation Gages
 - Gage 1
 - 01Jan2000, 00:00 - 05Jan2000, 00:00

Below the project tree, there are tabs for 'Components', 'Compute', and 'Results'. Under the 'Components' tab, there are sub-tabs for 'Reach', 'Routing' (circled in red), and 'Options'. The 'Routing' sub-tab is active, showing the following parameters for 'Basin 1' and 'Element Name: Reach-1':

- Initial Type: Discharge = Inflow (dropdown menu)
- *Muskingum K (HR): 2.7 (text input field)
- *Muskingum X: 0.1 (text input field)
- Subreaches: 1 (spin box)

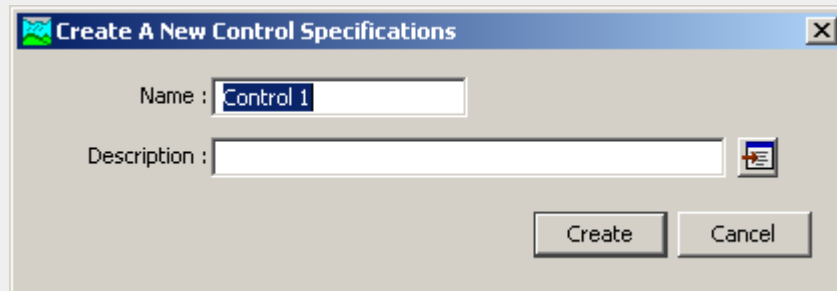
HEC-HMS: Running a simulation

Steps

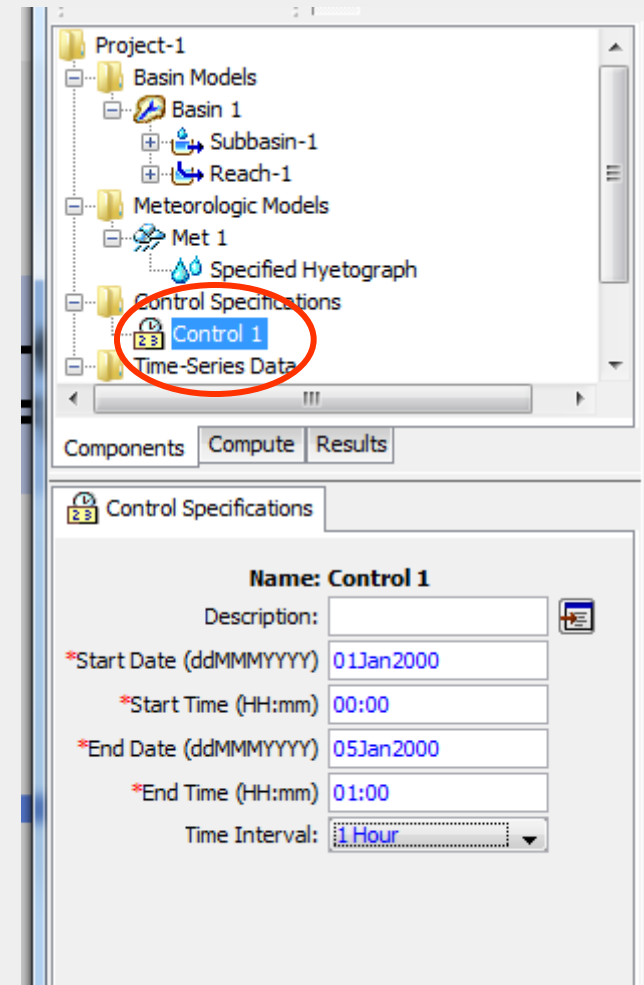
1. Create new Control Specification
2. Create Simulation Run
3. Compute the Simulation Run
4. View Results

HEC-HMS

- Components → Control Specifications Manager
→ New... → Create



- Control settings:
 - * Start Date: 01Jan2000
 - * Start Time: 00:00
 - * End Date: 05Jan2000
 - * End Time: 01:00
 - * Time Interval: 1 hour

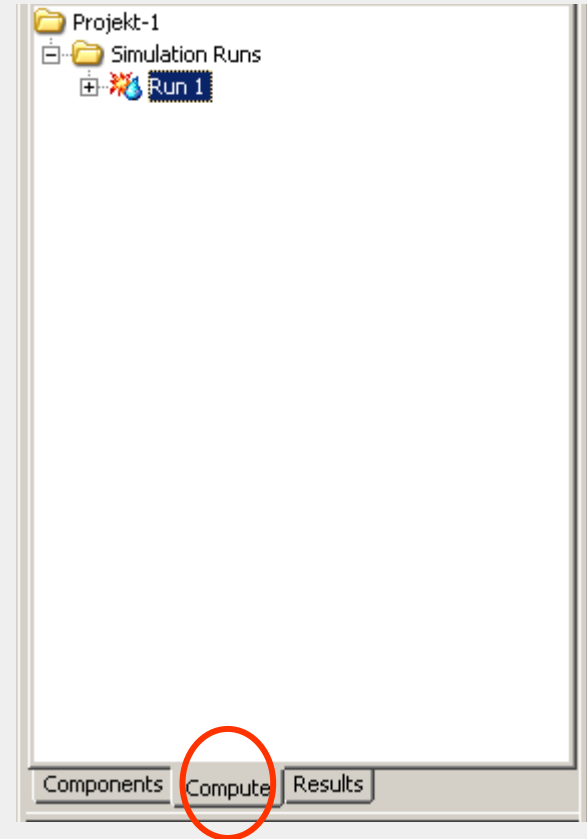


HEC-HMS

- Setting up a simulation run:
- Compute → Create Compute → Simulation Run
 - * Name: **Run 1**
 - * Basin Model: **Basin 1**
 - * Meteorological Model: **Met 1**
 - * Control Specification: **Control 1**

HEC-HMS

- in Watershed Explorer: Compute
- Run simulation:
 - * Compute Current Run



HEC-HMS: Viewing results

Steps

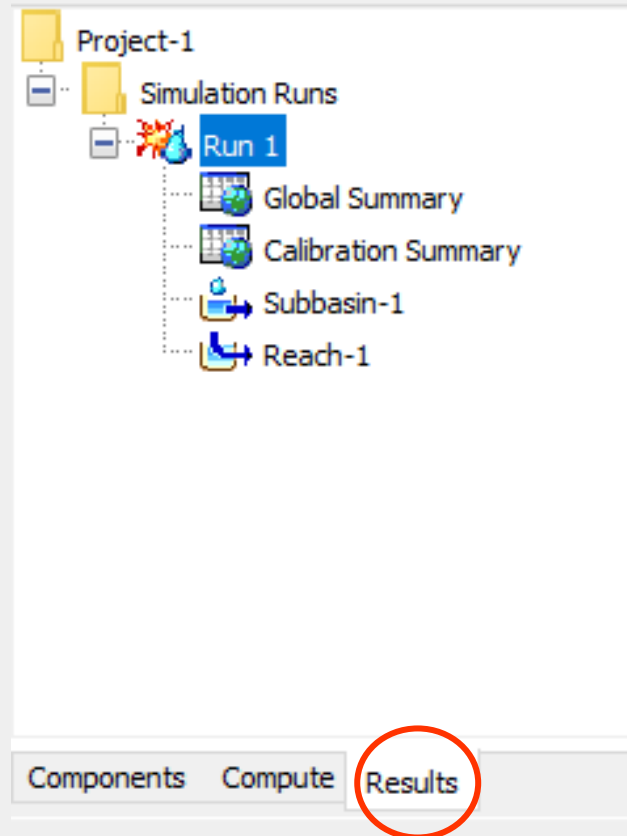
1. Global summary table
2. Element graph
3. Element summary table
4. Element time series
5. Spatial Results

How to access results

1. Watershed explorer *Results* tab
2. Right-click option in basin map
3. Compute toolbar
4. Results menu
5. Animation Toolbar

HEC-HMS

- Display simulation results



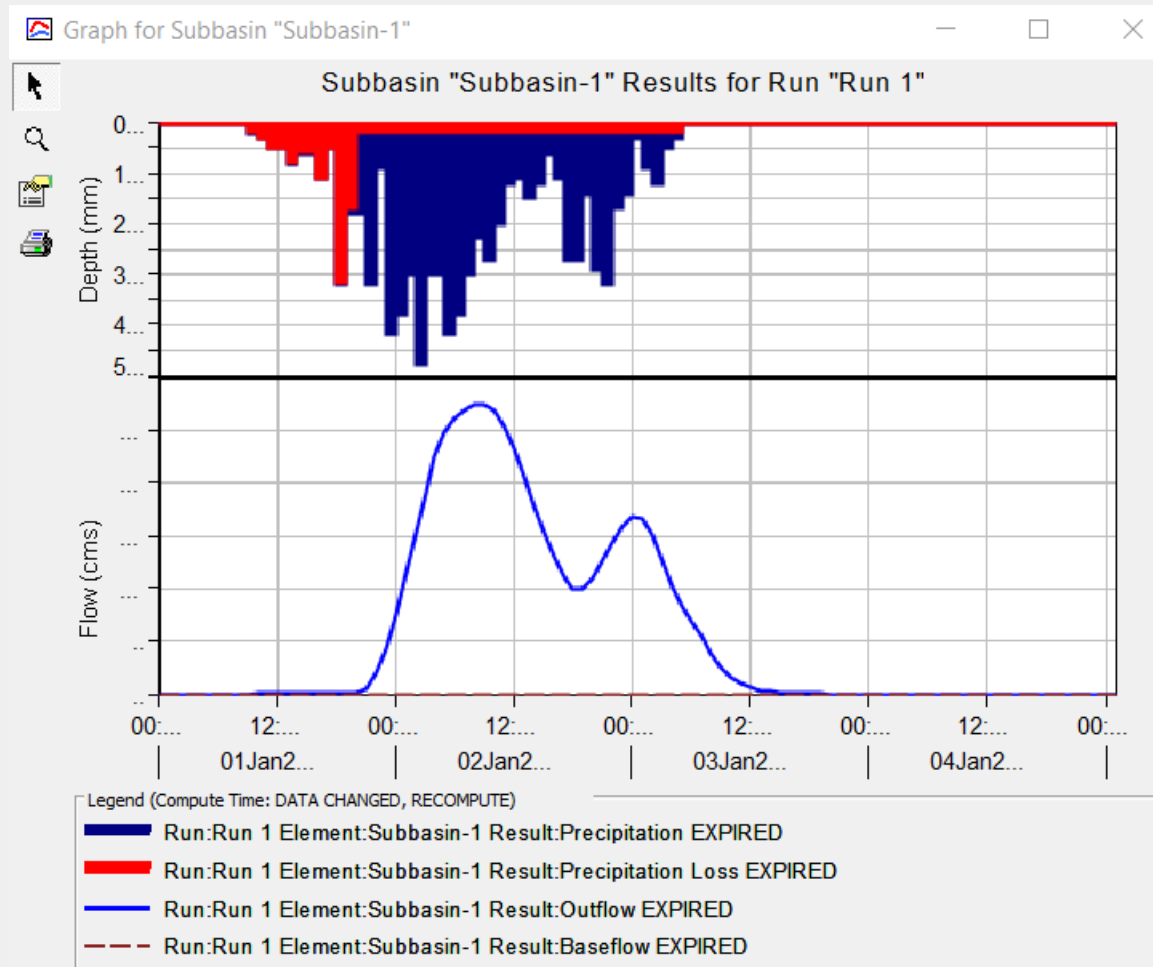
HEC-HMS

- Global Summary

Global Summary Results for Run "Run 1"				
Project: Project-1 Simulation Run: Run 1				
Start of Run: 01Jan2000, 00:00		Basin Model: Basin 1		
End of Run: 05Jan2000, 01:00		Meteorologic Model: Met 1		
Compute Time: DATA CHANGED, RECOMPUTE		Control Specifications: Control 1		
Show Elements:	All Elem...	Volume Units:	☒ MM ☐ 1000 M3	Sorting: Hydrol...
Hydrologic Element	Drainage Area (KM2)	Peak Discharge (M3/S)	Time of Peak	Volume (MM)
Subbasin-1	30	27.4	02Jan2000, 09:00	65.42
Reach-1	30	26.0	02Jan2000, 11:00	65.42

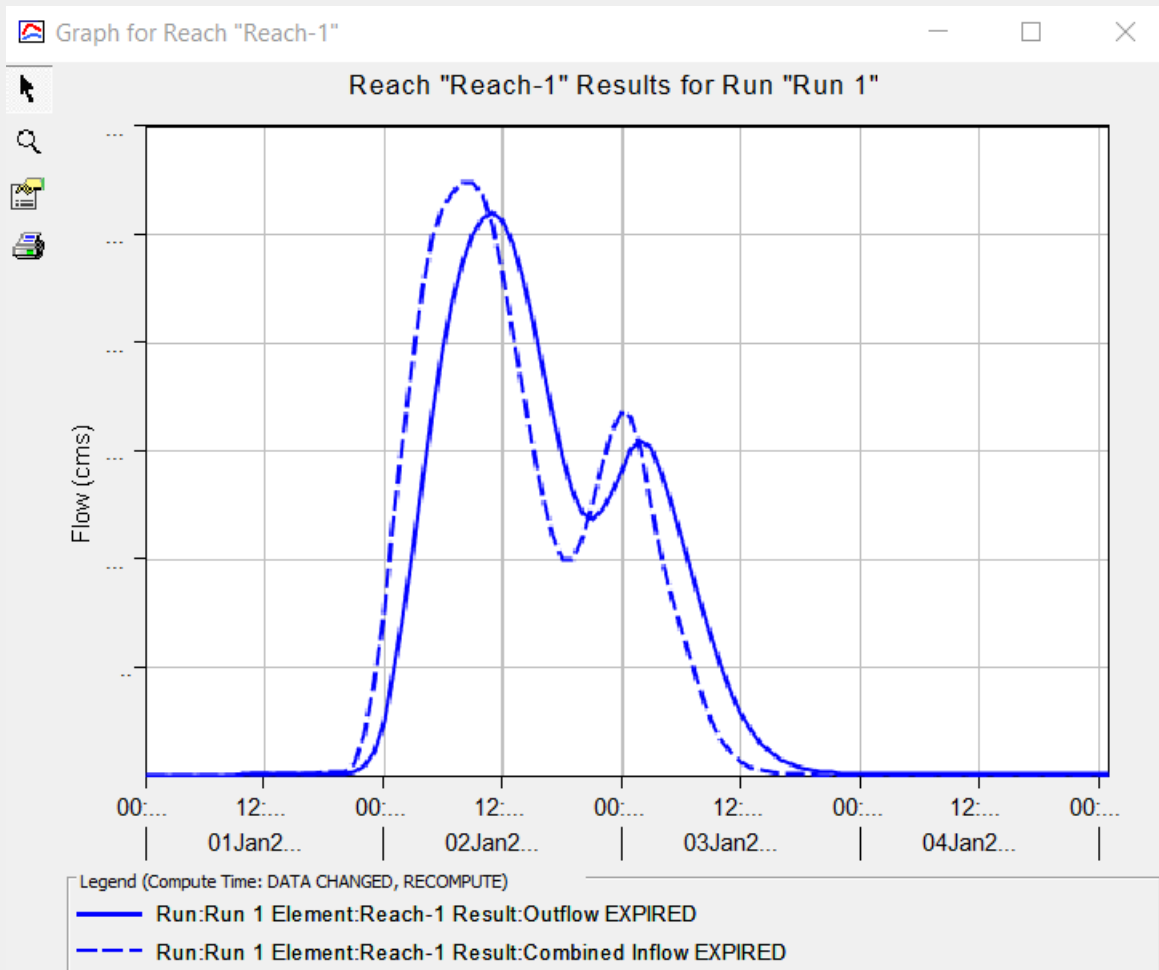
HEC-HMS

- Subbasin-1 → Graph



HEC-HMS

- Reach-1 → Graph
- Note: the time series can be exported as text file and then analysed further or plotted using e.g. Python or Excel



Explore

- Discuss the discharge response to the given rainfall hyetograph.
- Compare the duration of the rainfall event to the duration of the hydrograph (base time: time when the hydrograph starts and ends with a zero value).
- Determine the water budget for the subbasin in mm (how much water did it get from rainfall, how much was lost to infiltration, and how much came out as runoff); note that this is always a good first check to see if something might be wrong with the hydrological model setup.
- View other results such as cumulative precipitation, soil infiltration, excess precipitation, precipitation loss and direct runoff, and briefly describe what you understand by each of these.

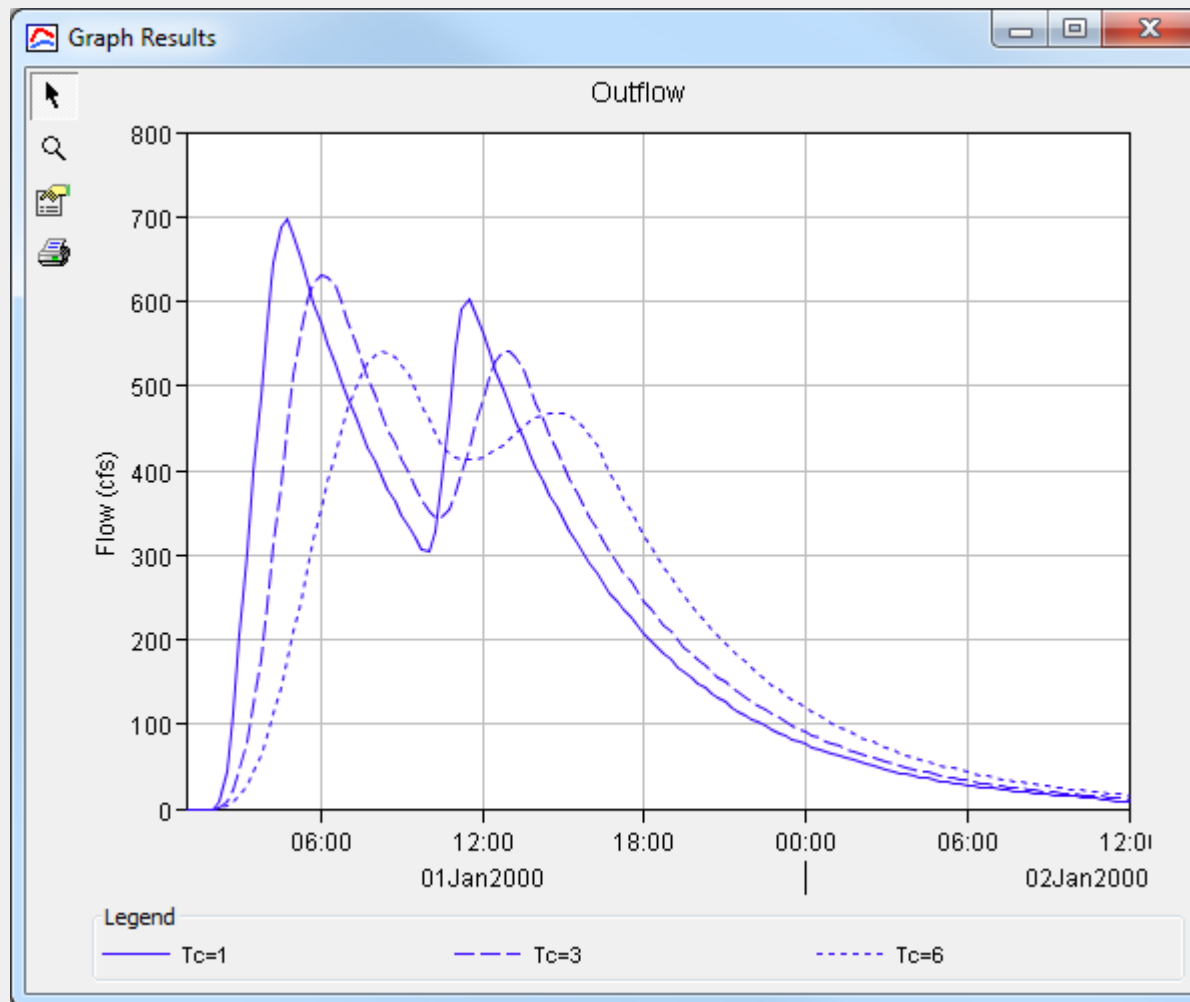
Explore

- Trial different rainfall events (different amounts and temporal distributions) and study how these are affecting the streamflow hydrograph. Specifically, look at the total amount of rainfall for each event, the shape of the rainfall hyetograph and the discharge hydrograph, and how the time to peak, basin time and base time vary for these events.
- For each event, report the total amount of rainfall in mm, peak discharge value in m^3/s , time to peak in hours, basin lag (time from rainfall peak to hydrograph peak) in hours and base time in hours of the output hydrograph.
- Trial different parameterisations and assess the sensitivity of the model output to parameter value changes (see assignment).

Check your understanding

- Explain the behaviour of the hydrograph due to the changes in the respective parameter (T_c and R) on the following two slides

Transform method Clark UH: Adjusting T_c



Transform method Clark UH: Adjusting R

